

Lecture 34: Bayesian Estimation - Some Terminologies and Definitions.

Before we consider a r.v. $X \sim p_\theta \in \mathcal{P} = \{p_\theta: \theta \in \Theta\}$, with at least one unknown parameter.

A decision ^{reg.} ~~an~~ estimator of θ is taken based on data X or $\{x_1, \dots, x_n\}$.

There is a loss $L(\theta, d(x_1, \dots, x_n))$ due to taking this decision d .

We look for an optimal d for minimizing the risk associated with decision, and define a risk for $R(\theta, d) = E(L(\theta, d(X)))$.
(Risk is expected loss).

The decision $d(x_1, \dots, x_n)$ is about finding a suitable (and accepting it as an) estimator of θ , say, $\delta(x_1, \dots, x_n)$. Thus for a sample space $\mathcal{X}$, we have a decision space $\mathcal{D}$.

Thus, in the case of frequentists/classical analysis we do not restrict us to the class of unbiased estimators, consider minimum of only variance, etc; and consider a general (usually, convex) loss function.

Another major shift is in the role of the parameter, which is no longer considered as an unknown, ~~fixed~~ fixed quantity. In Bayesian analysis the parameter θ is itself a r.v. having a (prior) distribution of its own, usually denoted by $\pi(\theta)$.

Distribution of X is thus a conditional distn $X|\theta$. ($f(x|\theta)$)

When data is observed we use this info' to modify the prior distn of θ by the posterior distn of $\theta|x=x$, say, $\pi(\theta|x)$.

As mentioned above, we look for an optimal decision $d^* \Rightarrow$

$$R(\theta, d^*) = E(L(\theta, d^*)) \leq R(\theta, d) = E(L(\theta, d)) \quad \forall \theta \in \Theta, d \in \mathcal{D}.$$

There are 2 approaches to achieve this.

I.

Consider the average/weighted average of risk of the decision/estimator $\delta(x)$ of θ , 2/34.

$$r(\pi, \delta) = \int R(\theta, \delta) \pi(\theta) d\theta, \quad \left[\text{If we add up to 1, } \int \pi(\theta) d\theta = 1 \right]$$

δ_{B_i} (say) minimizing $r(\pi, \delta)$ over all δ (with respect to a loss & prior) is called the Bayes estimator of θ and the corresponding $r(\pi, \delta_B)$ is the minimum average risk, is the Bayes' risk.

II. Consider the maximum risk, $\sup_{\theta} R(\theta, \delta) = r(\delta)$, say,
and ~~try to~~ minimize the maximum risk $r(\delta)$ over δ .
The resulting estimator is called a Minimax estimator (of θ).

We shall consider only Bayes estimator for the remaining part of the course.

So, the set-up is:

$X \sim f_{\theta} \in \Theta = \{f_{\theta} : \theta \in \Theta\}$, to estimate θ

$f_{\theta}(x)$ is now the conditional of $x|\theta$.

Assume $\theta \sim \pi(\theta)$, a prior distribution, completely specified (may be relaxed later on).

Loss $L(\theta, d)$ for estimating θ by $\delta(x_1, \dots, x_n)$, i.e. taking decision $d(x_1, \dots, x_n)$

Risk $R(\theta, \delta) = \text{expected loss} = E(L(\theta, \delta(X)))$.

δ_{π} is Bayes estimator of θ wrt the loss $L(\theta, d)$ & prior $\pi(\theta)$ if

for δ_{π} , $r(\pi, \delta_{\pi}) = \inf_{\delta} r(\pi, \delta)$, when $r(\pi, \delta) = \int R(\theta, \delta) \pi(\theta) d\theta$.

NOTE,

$$\begin{aligned} r(\pi, \delta) &= \int R(\theta, \delta) \pi(\theta) d\theta = \int \int L(\theta, \delta(x)) f(x|\theta) dx \pi(\theta) d\theta \\ &= \int \int L(\theta, \delta(x)) f(x|\theta) \pi(\theta) dx d\theta = \int \int L(\theta, \delta(x)) \pi(\theta|x) f_X(x) d\theta dx \\ &= \int E_{\theta|x} (L(\theta, \delta(x)) | x=x) f_X(x) dx. \end{aligned}$$

So, to minimize $r(\pi, \delta)$ over θ, δ involves minimizing $E_{\theta|x} (L(\theta, \delta(x)) | x=x)$, i.e. the risk of the posterior distn $\pi(\theta|x)$.

Now, if we consider the squander error loss fn, viz., $L(\theta, d) = (\theta - d)^2$,

$r(\pi, \delta) = \int \int (\theta - \delta(x))^2 \pi(\theta|x) d\theta f_X(x) dx$ is minimized

at $\delta(x) = E(\theta|x)$ and the Bayes estimator wrt the loss fn $L(\theta, d) = (\theta - d)^2$ & a prior π , is $\hat{\delta}_{\text{Bayes}} = E(\theta|x)$, i.e. Mean of the posterior distribution.